

4.7 Stanford Manipulator

I. Cinemática Direta (posição):

Dando $C_i = \cos(\theta_i)$ e $S_i = \sin(\theta_i)$, $i = \{1, 2, 4, 5\}$

$$x = C_1 S_2 d_3 - S_1 d_2 + d_6 (C_1 C_2 C_4 S_5 + C_1 C_5 S_2 - S_1 S_4 S_5)$$

$$y = S_1 S_2 d_3 + C_1 d_2 + d_6 (C_1 S_4 S_5 + C_2 C_4 S_1 S_5 + C_5 S_1 S_2)$$

$$z = C_2 d_3 + d_6 (C_2 C_5 - C_4 S_2 S_6)$$

II. Relação Cinemática Diferencial:

$$\dot{p} = J \dot{\theta}$$

Na qual, $\dot{p}^T = [\dot{x}, \dot{y}, \dot{z}]$ $\dot{\theta} = [\dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_4, \dot{\theta}_5]$

$$J = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial \theta_4} & \frac{\partial x}{\partial \theta_5} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial \theta_4} & \frac{\partial y}{\partial \theta_5} \\ \frac{\partial z}{\partial \theta_1} & \frac{\partial z}{\partial \theta_2} & \frac{\partial z}{\partial \theta_4} & \frac{\partial z}{\partial \theta_5} \end{bmatrix}$$

III. Cálculo das Derivadas Parciais:

Sabendo que

$$\frac{d(\cos(\theta))}{d\theta} = -\sin(\theta)$$

$$\frac{d(\sin(\theta))}{d\theta} = \cos(\theta)$$

Columna 1 $\left(\frac{\partial p}{\partial \theta_1} \right)$:

$$\frac{\partial x}{\partial \theta_1} = -s_1 s_2 d_3 - c_1 d_2 + d_6 (-s_1 c_2 c_4 s_5 - s_1 c_5 s_2 - c_1 s_4 s_5)$$

$$\frac{\partial y}{\partial \theta_1} = c_1 s_2 d_3 - s_1 d_2 + d_6 (-s_1 s_4 s_5 + c_2 c_4 c_1 s_5 + c_5 c_1 s_2)$$

$$\frac{\partial z}{\partial \theta_1} = 0$$

Columna 2 $\left(\frac{\partial p}{\partial \theta_2} \right)$:

$$\frac{\partial x}{\partial \theta_2} = c_1 c_2 d_3 + d_6 (-c_1 s_2 c_4 s_5 + c_1 c_5 c_2)$$

$$\frac{\partial y}{\partial \theta_2} = s_1 c_2 d_3 + d_6 (-s_2 c_4 s_1 s_5 + c_5 s_1 c_2)$$

$$\frac{\partial z}{\partial \theta_2} = -s_2 d_3 + d_6 (-s_2 c_5 - c_4 c_2 s_5)$$

Columna 3 $\left(\frac{\partial p}{\partial \theta_4} \right)$:

$$\frac{\partial x}{\partial \theta_4} = d_6 (-c_1 c_2 s_4 s_5 - s_1 c_4 s_5)$$

$$\frac{\partial y}{\partial \theta_4} = d_6 (c_1 c_4 s_5 - c_2 s_4 s_1 s_5)$$

$$\frac{\partial z}{\partial \theta_4} = d_6 s_4 s_2 s_5$$

Columna 4 $\left(\frac{\partial p}{\partial \theta_5} \right)$:

$$\frac{\partial x}{\partial \theta_5} = d_6 (c_1 c_2 c_4 c_5 - c_1 s_5 s_2 - s_1 s_4 c_5)$$

$$\frac{\partial y}{\partial \theta_5} = d_6 (c_1 s_4 c_5 + s_2 c_4 s_1 c_5 - s_5 s_1 s_2)$$

$$\frac{\partial z}{\partial \theta_5} = d_6 (-c_2 s_5 - c_4 s_2 c_5)$$

4.8 SCARA Manipulador

I. Cinemática Direta (posição)?

Dando $c_i = \cos(\theta_i)$ e $s_i = \sin(\theta_i)$, $i = \{1, 2\}$

$$x = a_3 c_1 + a_2 c_{12}$$

$$y = a_1 s_1 + a_2 s_{12}$$

$$z = -d_3 - d_4$$

II. Relação Cinemática Diferencial:

$$\dot{p} = J \dot{\theta}$$

No qual, $\dot{p}^T = [\dot{x}, \dot{y}, \dot{z}]$ $\dot{\theta} = [\dot{\theta}_1, \dot{\theta}_2]$

$$J = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \\ \frac{\partial z}{\partial \theta_1} & \frac{\partial z}{\partial \theta_2} \end{bmatrix}$$

III. Cálculo das Derivadas Parciais:

Sabendo que

$$\frac{d(\cos(a.b))}{da} = -a \cdot \sin(a.b)$$

$$\frac{d(\sin(a.b))}{da} = a \cos(a.b)$$

Column 1 $\left(\frac{\partial p}{\partial \theta_1} \right)$

$$\frac{\partial x}{\partial \theta_1} = -a_1 s_1 - a_2 \theta_2 s_{12}$$

$$\frac{\partial y}{\partial \theta_1} = a_1 c_1 + a_2 \theta_2 c_{12}$$

$$\frac{\partial z}{\partial \theta_1} = 0$$

Column 2 $\left(\frac{\partial p}{\partial \theta_2} \right)$

$$\frac{\partial x}{\partial \theta_2} = -a_2 \theta_1 s_{12}$$

$$\frac{\partial y}{\partial \theta_2} = a_2 \theta_1 c_{12}$$

$$\frac{\partial z}{\partial \theta_2} = 0$$

$$J = \begin{bmatrix} -a_1 s_1 - a_2 \theta_2 s_{12} & -a_2 \theta_1 s_{12} \\ a_1 c_1 + a_2 \theta_2 c_{12} & a_2 \theta_1 c_{12} \\ 0 & 0 \end{bmatrix}$$